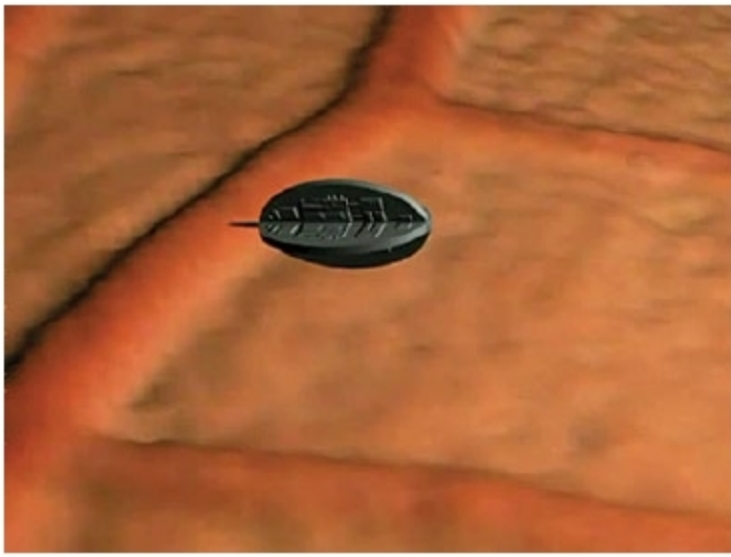




OctoMag microbot with a protruding needle to inject medicine into the exact affected body part (Credit: <https://gigazine.net>)

guided micro-bot called OctoMag, which is the size of a few human hair (285µm) and has redefined eye surgeries. OctoMag can be controlled magnetically to allow easy access and eliminate the necessity of slicing the eye open. The manipulating system consists of electromagnetic coils, which guide OctoMag for wireless eye surgery.

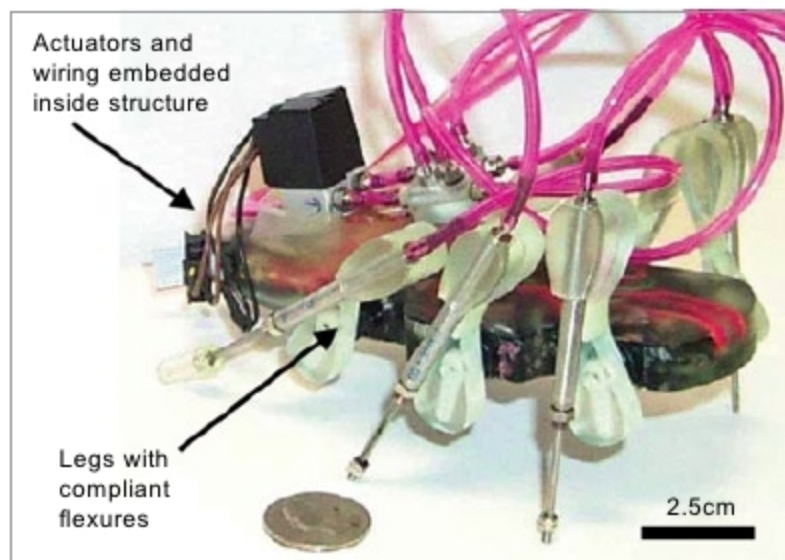
Micro-bots in space. Using micro-bots in space provides such advantages as decreased requirement of resources and the ability to replace larger and more discrete equipment, thus improving performance and efficiency and reducing cost, which are of main concern in space environment. Based on size and energy requirements, power sources used include coin cells, super capacitors and film batteries built using various polymer materials. Thin-film polymer batteries are more reliable than the rest in meeting power, space and operating temperature requirements. Other energy-scavenging methods include induction, microwave energy transmission and photovoltaics.

Communication methods for micro-bots in space include radio frequency (RF), coil induction and optical technologies. Walking, crawling, swimming or flying are the motions available for micro-bots in space. Flying abilities and aerodynamic features of several insects has provided inspiration for many concepts and inventions of tiny, winged micro-bots such as drones. Their capability to move easily through complex environments makes monitoring, surveillance, search, rescue and obtaining information of the environment easy.

Since digital devices may not respond in changeable situations, artificial neural networks inspired by neural networking within insects are used to solve the problem. Methods used include actuation by electrostatics, electromagnetic, lead-zirconium-titanate and shape memory alloys, which convert thermal energy to kinetic energy. Based on the natural movement of cockroaches, a miniature robot, Sprawlita (a dynamically-stable running hexapod), has been developed using pneumatic motors.

Piezoelectric actuators, which are scaled to small sizes and can be put into a variety of configurations, are used. Copying the way fish and sea insect move, micro-bots are built with aerial and aquatic locomotive abilities using movable, flexible wing designs.

Miniature robots in warfare. Progress in miniaturisation has helped create a new generation of micro-bots for use by the armed forces. These present new capabilities—easy to carry and capable of squeezing in undetected—into otherwise inaccessible spaces. These can be gainfully employed for providing life-saving intelligence, surveillance and reconnaissance (ISR) information to the operator, and relaying it to a communication, command and control centre. Swarming technology provides the ability to full-size robots and micro-bots to autonomously interact to carry out joint missions, responding to moves made as well as data captured by each other.



Sprawlita, a miniature robot used in space (Credit: <https://www.researchgate.net>)

Micro-bot systems are most effective in field operations for explosives reconnaissance and deactivation, to capture data on enemy positions and locate survivors of natural disasters due to their ability to fly, crawl, climb and wiggle into inaccessible spaces. For example, their mobility, durability, light weight and methods of locomotion can be combined to innovate a device that can be thrown over walls and in through windows, and be steered remotely to capture data using a range of onboard sensors accessories, cameras and other payloads.

Dumbbell-shaped XT is a miniature robot used with a handheld control unit to control and display the real-time video it captures. Together, these are difficult to detect and destroy by the enemy as they are small, operate quietly and are armed with an infrared (IR) optical system that can operate in low light, enabling the operator to operate in darkness. The system can be deployed within seconds, be thrown up as high as 36 metres, and transmits video up to 30 metres indoors and 91 metres outdoors.

Micro-bots are very useful in military, terrorism and natural disaster operations to perform tasks that are otherwise excessively dangerous or dirty for humans to carry out.

Challenges and perspectives

Micro-bots are fast becoming an important member of the robotic family that already includes industrial robots, service robots, medical and rehabilitation robots, cleaning robots, humanoid robots and others. The field of micro-robotics is making huge progress. Many technologies are under development

and integration to improve the device design and control, localisation, fabrication and power supply.

Major challenges in the development and commercialisation of micro-bots include innovating efficient remote power sourcing, improving effective and efficient locomotion mechanisms, and the ability to accurately maneuver them. Driving, controlling and propelling the devices